Cross-Flow Head-of-Line Blocking in Wi-Fi 7 Multi-Link Operation: Analytical Modeling and Sequence-Aware Scheduling

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Abstract—The IEEE 802.11be (Wi-Fi 7) amendment introduces Multi-Link Operation (MLO) to aggregate throughput and reduce latency across multiple radio bands (2.4 GHz, 5 GHz, and 6 GHz). However, under standard multi-link Block Acknowledgment (BA) agreements, all application flows mapped to a given Traffic Identifier (TID) share a single, unified 12-bit sequence-number space and a common receiver reordering buffer. When heterogeneous frames are scheduled across parallel radio links with asymmetric transmission delays, contention backoffs, and frame error rates, out-of-order physical arrivals create severe Head-of-Line (HOL) reordering stalls. Crucially, a delayed frame from a bulk background flow traps successfully received frames from latency-critical flows in the receiver buffer—a phenomenon we define as *Cross-Flow Head-of-Line (CF-HOL) Blocking*.

In this paper, we develop the first formal analytical framework and sender-side scheduling policy for CF-HOL blocking under standard IEEE 802.11be receiver semantics. First, we prove an orthogonal decomposition theorem demonstrating that total receiver reorder delay partitions additively and uniquely into intra-flow self-blocking and inter-flow cross-blocking ($\Delta_i = \mathcal{S}_i + \mathcal{X}_i$) at the buffer accounting boundary. Second, we establish the Multi-Link Sequence Divergence and Buffer Stability Bound ($\lambda_2 < (W - z_\alpha \sigma_{\Delta s})/\mu_1$), showing that exceeding the critical threshold $\lambda^* = W/\mu_1$ triggers severe window stalls and tail latency surge. Third, we formulate the joint packet selection, link mapping, and sequence assignment problem, proving that the optimal pairwise sequence order between concurrent frames is uniquely determined by a transcendental threshold root $u^*(w_a, w_b)$ solved via Newton-Raphson iteration. Fourth, we establish a conditional stochastic uncertainty lower bound demonstrating that concurrent multi-link dispatch over stochastic Gaussian channels cannot achieve zero cross-flow delay. Guided by these theoretical results, we design *Sequence-Aware Stochastic Partitioning (SASP)*, a transmitter-side Upper MAC scheduling algorithm that suppresses receiver stalls without modifying standard IEEE 802.11be receiver-side protocol semantics. Rigorous simulations across 30 random seeds demonstrate that SASP suppresses cross-flow blocking by 16.1%–17.7% across link asymmetry sweeps, reduces cross-blocking by up to 90.2% under near-saturation loads, suppresses high-loss cross-blocking by 36.2%, and delivers a 20.5% reduction in priority-weighted cost in algorithmic ablation while preserving 100% link goodput and executing in $\leq 15.3\,\mu$s per decision.

Index Terms—Wi-Fi 7, IEEE 802.11be, Multi-Link Operation (MLO), Block Acknowledgment, Head-of-Line Blocking, Packet Reordering, Stochastic Scheduling, Upper MAC.

1 Introduction

The IEEE 802.11be Extremely High Throughput (EHT) amendment, commercialized under the trade name Wi-Fi 7, establishes Multi-Link Operation (MLO) as its architectural cornerstone [1]. By allowing affiliated Multi-Link Devices (MLDs) to coordinate transmissions across multiple physical interfaces operating simultaneously in the 2.4 GHz, 5 GHz, and 6 GHz bands, MLO delivers unprecedented multi-gigabit throughput and promises deterministic sub-millisecond latencies for emerging real-time applications such as cloud gaming, augmented reality (XR), and industrial robotics.

To manage parallel radio access without redesigning higher network layers, the IEEE 802.11be MAC architecture is divided into an Upper MAC (U-MAC) and multiple Lower MACs (L-MACs) [2]. The U-MAC is responsible for queue management, multi-link packet scheduling, and flow-level coordination, whereas each L-MAC manages link-specific physical carrier sensing, Enhanced Distributed Channel Access (EDCA) backoff, and Physical Layer Convergence Protocol (PLCP) encapsulation.

Under the 802.11be specification, the Block Acknowledgment (BA) mechanism operates at the U-MAC level per peer MLD and Traffic Identifier (TID $\in \{0, \ldots, 7\}$) [1]:

1) A single, unified 12-bit sequence-number space ($s \in [0, 4095]$) is shared by all MAC Service Data Units (MSDUs) transmitted under a given TID across all active links.
2) The receiver MLD maintains a single sliding receive reordering buffer governed by window size $W \in \{64, 256, 1024\}$ and window base W_{start}.
3) The receiver delivers MSDUs to the upper layers (Logical Link Control/IP) strictly in monotonically increasing sequence-number order.

While this unified sequence space guarantees backward compatibility with legacy 802.11 state machines and prevents out-of-order segment delivery to transport protocols like TCP, it creates an acute performance vulnerability when links exhibit heterogeneous physical channel characteristics [3], [4]. In modern wireless environments, parallel links inevitably differ in bandwidth (e.g., 20 MHz on 2.4 GHz

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vs. 160/320 MHz on 5/6 GHz), PHY data rates, channel contention (EDCA contention slot delays), and Frame Error Rates (FER).

When sequence numbers are assigned conventionally in arrival order (FIFO), an earlier-sequenced frame dispatched over a slower, congested, or lossy link will complete physical transmission significantly later than subsequently-sequenced frames dispatched over a clear, high-speed link. Consequently, successfully received frames are trapped inside the receiver reordering buffer awaiting the missing earlier sequence number.

This bottleneck becomes critical when heterogeneous application flows share the same TID. Because the IEEE 802.11 standard defines only eight TIDs mapped to four Access Categories (AC_VO, AC_VI, AC_BE, AC_BK), host operating systems routinely multiplex diverse application flows—such as interactive VoIP, cloud gaming telemetry, and bulk background file synchronization—into the default Best Effort Access Category (TID 0) [3], [5]. In such deployments:

- A bulk packet dispatched on a slow 2.4 GHz link receives sequence number s.
- Latency-critical gaming packets dispatched on a 6 GHz link receive sequence numbers $s+1, s+2, \ldots, s+k$.
- Although the gaming packets arrive physically in fractions of a millisecond, the receiver cannot deliver them to the host network stack until packet s arrives or undergoes retries.

We formally define this phenomenon as **Cross-Flow Head-of-Line (CF-HOL) Blocking**.

Recent empirical measurements by Oliveira et al. [4] in ns-3 confirmed that multi-link reordering causes severe latency degradation. However, existing work suffers from three decisive limitations:

1) *Absence of Mathematical Modeling:* Prior studies [4], [6] treat multi-link reordering empirically without formalizing the stochastic coupling between channel completion variances and sliding BA window dynamics.
2) *Confounded Delay Metrics:* Existing literature measures only aggregate end-to-end latency, failing to mathematically decouple intra-flow self-reordering from inter-flow cross-blocking.
3) *Non-Standard or Receiver-Side Proposals:* Proprietary patents, such as MediaTek US12470496B2 [7], propose receiver-side Deep Packet Inspection (DPI) to construct per-flow reorder queues. This violates standard IEEE 802.11be protocol semantics, requires specialized receiver silicon, and leaves commercial off-the-shelf client devices completely unprotected.

Contributions: In this paper, we resolve these challenges by establishing the first mathematically rigorous analytical model and sender-side scheduling architecture for CF-HOL blocking in Wi-Fi 7:

- **Orthogonal Delay Decomposition (Theorem 1):** We prove that total receiver reordering delay decomposes additively, uniquely, and non-negatively into intra-flow self-blocking and inter-flow cross-blocking ($\Delta_i = \mathcal{S}_i + \mathcal{X}_i$), providing an exact in-buffer accounting metric.
- **Sequence Divergence & Stability Bound (Theorem 2):** We derive the stochastic divergence condition $\lambda_2 < (W - z_\alpha \sigma_{\Delta s})/\mu_1$ and asymptotic collapse boundary $\lambda^* = W/\mu_1$, showing that exceeding the threshold induces severe sliding window stalls with quadratic delay scaling $\frac{1}{2}\lambda_2 \delta^2$.
- **Optimal Pairwise Predictive Sequencing Rule (Theorem 3):** We formulate the joint scheduling and sequence assignment problem, proving that the optimal causal sequence order between concurrent candidate frames is uniquely determined by a transcendental threshold root $u^*(w_a, w_b)$ solved via Newton-Raphson in ≤ 4 steps.
- **Conditional Uncertainty Floor (Theorem 4):** We establish a fundamental lower bound proving that concurrent multi-link dispatch over stochastic Gaussian links cannot achieve zero cross-flow delay, while explicitly delineating its boundary conditions (e.g., Single-Link Operation achieves $\mathcal{X} = 0$).
- **The SASP Algorithm & Comprehensive Validation:** We design *Sequence-Aware Stochastic Partitioning (SASP)*, an Upper MAC algorithm executing in $\leq 15.3\,\mu$s in Python ($< 4\,\mu$s typically). Benchmarked across eight baselines and 30 random seeds, SASP reduces cross-flow blocking by 16.1%–17.7% under asymmetry, up to 90.2% under buffer load, and 36.2% under loss, while strictly adhering to standard 802.11be receiver semantics.

2 Related Work and State of the Art

2.1 Multi-Link Operation in IEEE 802.11be

The Multi-Link Operation (MLO) architecture defined in IEEE Std 802.11be-2024 provides two primary operational modes: Simultaneous Transmit and Receive (STR) and Non-Simultaneous Transmit and Receive (NSTR) [1]. In STR mode, an MLD can transmit or receive simultaneously on multiple links without cross-band in-device interference (e.g., between 5 GHz and 6 GHz). Carreño et al. [3] conducted early throughput studies of MLO, observing that when links exhibit severe rate asymmetry, naive multi-link aggregation can paradoxically degrade overall performance compared to single-link operation. Levy et al. [8] and Al-Hraishawi et al. [6] analyzed MLO delay under finite queue loads, but both studies assumed single-flow Poisson arrivals and ideal in-order reception.

2.2 Reorder Buffering and Block Acknowledgment Dynamics

The Block Acknowledgment (BA) mechanism originated in IEEE 802.11e and was extended in 802.11n/ac/ax for high-speed frame aggregation (A-MPDU). In 802.11be MLO, the BA agreement operates at the peer MLD level across all links affiliated with a TID [1].

Oliveira, Raposo, and Luís (2026) [4] provided the first in-depth empirical simulation of BA window stalls in ns-3, demonstrating that out-of-order frame arrivals cause

substantial buffer occupancy and delay spikes for UDP traffic. Sangeetha et al. [5] presented an analytical model for unsaturated MLO queues using a multi-dimensional Markov chain, but their formulation assumed that each traffic flow possesses a dedicated TID, sidestepping single-TID multi-flow contention. Choi, Jung, and Kim [2] explored joint L-MAC and U-MAC packet scheduling for NSTR MLO, focusing on power consumption and channel switching overhead rather than sequence number assignment.

2.3 Multipath Transport Schedulers vs. Wireless MAC

Out-of-order packet reordering has been extensively studied at the transport layer, particularly in Multipath TCP (MPTCP) [9]. Schedulers such as BLEST (Blocking Estimation-based MPTCP Scheduler) [9] predict head-of-line receiver blocking by comparing estimated path Round-Trip Times (RTTs).

However, transport-layer schedulers operate on continuous byte streams over long RTT timescales (10–100 ms) with explicit end-to-end ACK feedback. In contrast, wireless Upper MAC scheduling operates on discrete frames at sub-millisecond timescales ($100\,\mu$s–5 ms) under non-stationary EDCA contention backoff, small sliding sequence windows ($W \in \{64, 256\}$), and rapid burst fading. Consequently, MPTCP heuristics cannot be directly applied to the 802.11be U-MAC.

2.4 Architectural Boundary and Patent Landscape

To mitigate MLO reordering stalls, proprietary patent filings have emerged. Most notably, MediaTek patent US12470496B2 [7] proposes a *receiving apparatus* that performs Deep Packet Inspection (DPI) on incoming MPDUs, inspects IP/UDP 5-tuple headers, and demultiplexes packets into separate per-flow receive queues before releasing them to upper layers.

While effective in closed ecosystems, this approach has severe practical drawbacks:

1) It violates standard IEEE 802.11be receiver protocol semantics, which dictate a single TID reordering buffer.
2) It cannot accelerate standard commercial client stations (smartphones, IoT devices) that implement unmodified 802.11be MAC firmware.
3) It requires expensive packet header inspection hardware on power-constrained mobile receivers.

In contrast, our proposed SASP algorithm resides **strictly at the transmitting U-MAC**. The transmitter optimizes link dispatch and sequence numbering such that packets arrive physically in sequential order at the receiver. Standard commercial client devices receive packets within valid sequence windows and deliver them with zero stalls, guaranteeing full protocol compliance and complete independence from receiver-side patents.

3 System Model and Problem Formulation

3.1 Multi-Link MAC Sublayer Architecture

We consider an IEEE 802.11be Access Point (AP) MLD communicating with an affiliated non-AP station MLD over a set of $K \geq 2$ orthogonal radio links $\mathcal{K} = \{1, 2, \ldots, K\}$. We assume Simultaneous Transmit and Receive (STR) operation across non-overlapping frequency bands (Link 1: 2.4 GHz, Link 2: 5 GHz, Link 3: 6 GHz).

A set of $M \geq 2$ distinct application flows $\mathcal{F} = \{1, \ldots, M\}$ generate packets mapped to a single TID (e.g., TID 0, Best Effort). Each flow $f \in \mathcal{F}$ is characterized by an arrival rate λ_f and a delay penalty weight $w_f > 0$. Higher weights denote latency-sensitive traffic (e.g., $w_{\text{gaming}} = 20, w_{\text{bulk}} = 1$). Incoming packets are queued in flow-specific U-MAC buffers $Q_1, \ldots, Q_M$.

3.2 Lower MAC Channel Access and Service Times

Each radio link $k \in \mathcal{K}$ executes an independent EDCA channel access engine with parameters $(\text{AIFS}_k, \text{CW}_{\min,k}, \text{CW}_{\max,k})$ and PHY transmission rate R_k. When packet i of length L_i bits is dispatched to link $\ell_i = k$ at dispatch time t_i^{disp}, its total completion duration until successful Block Ack confirmation is:

$$D_i(k) = \sum_{r=1}^{N_{i,k}} \left(B_{k,r} + T_{\text{frame},k}(L_i)\right) + T_{\text{ack},k} \tag{1}$$

where $N_{i,k} \in [1, N_{\max}]$ is the number of transmission attempts until success, $B_{k,r}$ is the stochastic EDCA backoff slot duration, $T_{\text{frame},k}$ is the PLCP frame transmission time, and $T_{\text{ack},k}$ is the turnaround and Block Ack response time. The physical arrival time of packet i at the receiver is:

$$A_i = t_i^{\text{disp}} + D_i(\ell_i) \tag{2}$$

The link service duration $D_i(k)$ is modeled as a continuous random variable with mean μ_k, variance σ_k^2, and sub-Gaussian parameter ν_k^2. Channel error dynamics are captured via a two-state Gilbert-Elliott model alternating between Good (low FER) and Bad (high FER) channel states.

3.3 Receiver Block Ack Reorder Buffer

The receiver MLD manages a single reordering buffer of size $W \in \{64, 256, 1024\}$ for the given TID. The active sequence window is $[W_{\text{start}}, W_{\text{start}} + W - 1]$. Upon physical arrival of packet i at time A_i with sequence number s_i:

- If $s_i = W_{\text{start}}$: Packet i is immediately delivered to upper layers, and W_{start} advances to the lowest missing sequence number.
- If $W_{\text{start}} < s_i < W_{\text{start}} + W$: Packet i is retained in the buffer awaiting all missing predecessors $\{s' : W_{\text{start}} \leq s' < s_i\}$.
- If $s_i \geq W_{\text{start}} + W$: The buffer window overflows. To prevent indefinite stalls, W_{start} is forcefully advanced to $s_i - W + 1$, and all missing packets prior to the new W_{start} are declared lost.

Under reliable transmission, packet i is released to the host stack at time:

$$R_i = \max\left(A_i, \max_{k: s_k < s_i} A_k\right) \tag{3}$$

The total reordering delay experienced by packet i is $\Delta_i = R_i - A_i \geq 0$.

TABLE 1: System Model and Simulation Parameters

Parameter	Link 1 (2.4 GHz)	Link 2 (5 GHz)	Link 3 (6 GHz)
Channel Width	20 MHz	80 MHz	160 MHz
Base MCS	MCS 4	MCS 9	MCS 11
PHY Rate (R_k)	39.0 Mbps	480.4 Mbps	1201.0 Mbps
AIFS	3 slots	2 slots	2 slots
$\mathrm{CW}_{\min}/\mathrm{CW}_{\max}$	15 / 1023	15 / 1023	15 / 1023
Slot Time	9 μs	9 μs	9 μs
Base Delay (μ_k)	1.20 ms	0.40 ms	0.25 ms
Jitter (σ_k)	0.35 ms	0.12 ms	0.08 ms
Frame Error Rate	$0.05 - 0.15$	$0.02 - 0.08$	$0.01 - 0.05$

4 Theoretical Analysis and Theorems

4.1 Orthogonal Delay Decomposition

To rigorously study multi-flow interference, we must isolate the delay intrinsic to a flow's own dispersion from the delay imposed by competing flows sharing the TID.

Theorem 1 (Orthogonal Delay Decomposition)**.** *Let packet i belong to flow f_i with physical arrival time A_i and assigned sequence number s_i. Define the intra-flow completion boundary:*

$$R_i^{intra} \triangleq \max\left(A_i, \max_{j:f_j=f_i, s_j<s_i} A_j\right) \tag{4}$$

Then the total reordering delay $\Delta_i = R_i - A_i$ admits an exact, non-negative, orthogonal decomposition:

$$\Delta_i = \mathcal{S}_i + \mathcal{X}_i \tag{5}$$

where:

$$\mathcal{S}_i \triangleq R_i^{intra} - A_i = \max\left(0, \max_{j:f_j=f_i, s_j<s_i} A_j - A_i\right) \tag{6}$$

*denotes the intra-flow **Self-Blocking Delay**, and:*

$$\mathcal{X}_i \triangleq R_i - R_i^{intra} = \max\left(0, \max_{k:f_k\neq f_i, s_k<s_i} A_k - R_i^{intra}\right) \tag{7}$$

*denotes the inter-flow **Cross-Flow Blocking Delay**.*

Proof. By IEEE 802.11be in-order delivery semantics, the delivery epoch is $R_i = \max(A_i, \max_{k:s_k<s_i} A_k)$. The set of causal predecessors $\mathcal{P}_i = \{k : s_k < s_i\}$ partitions into disjoint intra-flow predecessors $\mathcal{P}_i^{\text{intra}} = \{j : f_j = f_i, s_j < s_i\}$ and inter-flow predecessors $\mathcal{P}_i^{\text{inter}} = \{k : f_k \neq f_i, s_k < s_i\}$. Hence:

$$R_i = \max\left(R_i^{\text{intra}}, \max_{k\in\mathcal{P}_i^{\text{inter}}} A_k\right) \tag{8}$$

Since $R_i \geq R_i^{\text{intra}} \geq A_i$, both $\mathcal{S}_i \geq 0$ and $\mathcal{X}_i \geq 0$. Summing:

$$\mathcal{S}_i + \mathcal{X}_i = (R_i^{\text{intra}} - A_i) + (R_i - R_i^{\text{intra}}) = R_i - A_i = \Delta_i \tag{9}$$

□

Lemma 1 (Attribution Invariance and Boundary Conditions)**.** *On an observed trace with fixed physical arrival epochs $\{A_i\}$ and sequence numbers $\{s_i\}$, the self-blocking delay $\mathcal{S}_i$ is strictly invariant to the arrival epochs, sequence numbers, and presence of all competing flows $\mathcal{F} \setminus \{f_i\}$.*

Physical Boundary Note: *This invariance is a buffer-level accounting property. In a real shared wireless medium, the physical presence of competing traffic introduces EDCA channel contention, collision-induced backoff escalation, and queuing delay, which dynamically alters $\{A_i\}$. Theorem 1 thus serves as an exact diagnostic metric for isolating in-buffer reordering stalls from channel service time dispersion.*

Proof. By definition, R_i^{intra} depends exclusively on A_i and the set $\{A_j : f_j = f_i, s_j < s_i\}$. Removing all frames of other flows leaves $\mathcal{P}_i^{\text{inter}} = \emptyset$, which forces $R_i = R_i^{\text{intra}}$ and $\mathcal{X}_i = 0$, while $\mathcal{S}_i$ remains unchanged. □

4.2 Multi-Link Sequence Divergence and Buffer Stability

We now analyze the stochastic behavior of the receiver sliding window when flows with asymmetric rates and delays share a TID under standard FIFO sequence assignment.

Theorem 2 (Sequence Divergence and Buffer Stability Bound)**.** *Consider an MLD transmitting bulk flow A on slow Link 1 (μ_1, σ_1^2) and interactive flow B on fast Link 2 (μ_2, σ_2^2) with mean asymmetry $\delta = \mu_1 - \mu_2 > 0$ and negotiated Block Ack window W. Under FIFO sequencing:*

1) *During the completion duration of a frame on Link 1, the sequence divergence Δs on Link 2 has mean $\mathbb{E}[\Delta s] = \lambda_B\mu_1$ and variance $\mathrm{Var}(\Delta s) = \lambda_B\mu_1 + \lambda_B^2\sigma_1^2$.*
2) *The cumulative cross-flow blocking duration per frame of flow A scales quadratically as:*

$$\mathbb{E}\left[\sum_{j=1}^{\Delta s} \mathcal{X}_{B,j}\right] = \frac{1}{2}\lambda_B\delta^2 \tag{10}$$

3) *(Buffer Stability Condition): To prevent sequence divergence from exceeding window W with probability $1-\alpha$ under IEEE 802.11be forced-advance semantics, the arrival rate must satisfy:*

$$\lambda_B < \lambda_{crit} \triangleq \frac{W - z_\alpha\sqrt{\mathrm{Var}(\Delta s)}}{\mu_1} \tag{11}$$

where $z_\alpha = \Phi^{-1}(1-\alpha)$. When $\lambda_B \geq \lambda^* \triangleq W/\mu_1$, the receiver buffer is saturated almost surely, triggering window stalls and forced advances.

Proof. Frame a of flow A is dispatched to Link 1 at $t = 0$, completing at D_1 with $\mathbb{E}[D_1] = \mu_1$. Concurrently, frames b_j $(j = 1, 2, \dots)$ are dispatched on Link 2 at rate λ_B with sequence numbers $s_j = s_a + j$. Frame b_j completes at $A_{b_j} = j/\lambda_B + D_{2,j}$. Whenever $j/\lambda_B + D_{2,j} < D_1$, packet b_j arrives before frame a and is cross-blocked: $\mathcal{X}_{b_j} = \delta - j/\lambda_B$. Integrating across $j \in [0, \lambda_B\delta]$:

$$\int_0^{\lambda_B\delta}\left(\delta - \frac{x}{\lambda_B}\right)dx = \lambda_B\delta^2 - \frac{\lambda_B^2\delta^2}{2\lambda_B} = \frac{1}{2}\lambda_B\delta^2 \tag{12}$$

By the law of total expectation and variance for compounded Poisson arrivals: $\mathbb{E}[\Delta s] = \lambda_B \mu_1$, $\mathrm{Var}(\Delta s) = \lambda_B \mu_1 + \lambda_B^2 \sigma_1^2$. In standard IEEE 802.11be (§ 10.24.7), when sequence divergence exceeds W, W_{start} force-advances. Requiring $\mathbb{P}(\Delta s \geq W) \leq \alpha$ yields $\mathbb{E}[\Delta s] + z_\alpha \sigma_{\Delta s} < W$. Solving for λ_B yields the stability condition. □

4.3 Optimal Predictive Sequencing Rule (SASP)

We now address the core scheduling problem: given candidate frames ready for transmission across parallel links, how should sequence numbers be assigned to minimize total weighted cross-flow blocking?

Theorem 3 (SASP Optimal Pairwise Threshold Rule)**.** *Let candidate frames a (flow f_a, weight w_a) and b (flow f_b, weight w_b, $f_a \neq f_b$) have completion times $A_a \sim \mathcal{N}(\mu_a, \sigma_a^2)$ and $A_b \sim \mathcal{N}(\mu_b, \sigma_b^2)$. Let $Z \triangleq A_a - A_b \sim \mathcal{N}(\mu_Z, \sigma_Z^2)$, with $\mu_Z = \mu_a - \mu_b$ and $\sigma_Z = \sqrt{\sigma_a^2 + \sigma_b^2}$.*

The sequence assignment decision minimizing expected weighted cross-flow blocking cost:

$$\begin{aligned} C(s_a < s_b) &= w_b \mathbb{E}[(A_a - A_b)^+], \\ C(s_b < s_a) &= w_a \mathbb{E}[(A_b - A_a)^+] \end{aligned} \tag{13}$$

is given by the strictly optimal pairwise threshold rule:

$$s_a < s_b \iff \frac{\mu_a - \mu_b}{\sqrt{\sigma_a^2 + \sigma_b^2}} \leq u^*(w_a, w_b) \tag{14}$$

where u^* is the unique root of the strictly monotonically increasing function:

$$h(u) \triangleq (w_b - w_a)\phi(u) + [w_a(1 - \Phi(u)) + w_b \Phi(u)]\, u = 0 \tag{15}$$

and $\phi(\cdot)$, $\Phi(\cdot)$ are the standard Gaussian PDF and CDF.

Properties of the Root u^*:

1) *When flow weights are equal ($w_a = w_b$), $u^* = 0$, and optimal sequence assignment is sorted strictly by expected completion time: $s_a < s_b \iff \mu_a \leq \mu_b$.*
2) *When flow b has higher priority ($w_b > w_a$), $u^* < 0$, satisfying $|u^*| \leq \sqrt{2 \ln(w_b/w_a)}$.*
3) *u^* is transcendental and computed via Newton-Raphson iteration in ≤ 4 steps.*
4) *For $N > 2$ candidate frames, pairwise sorting provides an $\mathcal{O}(M \log M)$ index heuristic whose multi-dimensional maximum is governed by Clark's order-statistics approximation.*

Proof. Using the identity $Z^+ - (-Z)^+ = Z$, we have $\mathbb{E}[(-Z)^+] = \mathbb{E}[Z^+] - \mu_Z$. The condition $C(s_a < s_b) \leq C(s_b < s_a)$ is equivalent to $(w_b - w_a)\mathbb{E}[Z^+] + w_a \mu_Z \leq 0$. For $Z \sim \mathcal{N}(\mu_Z, \sigma_Z^2)$, normalized variable $u = \mu_Z / \sigma_Z$. Truncated normal expectation yields:

$$\mathbb{E}[Z^+] = \sigma_Z \left[\phi(u) + u\Phi(u)\right] \tag{16}$$

Dividing by $\sigma_Z > 0$ yields:

$$(w_b - w_a)\phi(u) + [w_a(1 - \Phi(u)) + w_b \Phi(u)]\, u \leq 0 \tag{17}$$

Taking the first derivative of $h(u)$ with respect to u:

$$\begin{aligned} h'(u) = (w_b - w_a)(-u\phi(u)) + (w_b - w_a)(u\phi(u)) \\ + [w_a(1 - \Phi(u)) + w_b \Phi(u)] \end{aligned} \tag{18}$$

The first two terms cancel identically! Hence:

$$h'(u) = w_a(1 - \Phi(u)) + w_b \Phi(u) > 0, \quad \forall u \in \mathbb{R} \tag{19}$$

Because $h'(u) > 0$ strictly everywhere, $h(u)$ is strictly monotonically increasing from $-\infty$ to $+\infty$. By the Intermediate Value Theorem, $h(u) = 0$ possesses exactly one real root u^*. □

4.4 Conditional Stochastic Uncertainty Lower Bound

Theorem 4 (Conditional Stochastic Uncertainty Lower Bound)**.** *For any causal online scheduling policy $\pi \in \Pi$, conditional on candidate frames a and b being dispatched concurrently on disparate stochastic links with Gaussian completion variances $\sigma_a^2, \sigma_b^2 > 0$, the expected weighted cross-flow blocking is strictly lower-bounded by:*

$$\begin{aligned} \min\left(C(s_a < s_b), C(s_b < s_a)\right) \geq \\ \min(w_a, w_b) \frac{\sigma_Z}{\sqrt{2\pi}} \exp\left(-\frac{(\mu_a - \mu_b)^2}{2\sigma_Z^2}\right) \end{aligned} \tag{20}$$

where $\sigma_Z = \sqrt{\sigma_a^2 + \sigma_b^2}$. On symmetric links ($\mu_a = \mu_b$), this reduces to $\min(w_a, w_b)\frac{\sigma_Z}{\sqrt{2\pi}}$.

Falsification Boundaries: *The bound holds strictly conditional on concurrent multi-link dispatch over stochastic channels. Under Single-Link Operation (SLO) or deterministic TDMA, frames arrive in sequential order, achieving $\mathcal{X}_i = 0$ identically.*

Proof. For any sequencing decision $\omega \in \{s_a < s_b, s_b < s_a\}$, the cost is bounded by $\min(w_a, w_b)\min(\mathbb{E}[Z^+], \mathbb{E}[(-Z)^+])$. Applying the Gaussian lower bound $\min(\mathbb{E}[Z^+], \mathbb{E}[(-Z)^+]) \geq \sigma_Z \phi(u) = \frac{\sigma_Z}{\sqrt{2\pi}} e^{-u^2/2}$ completes the proof. □

4.5 Theoretical Failure Modes and Counterexamples

To illustrate the necessity of Theorem 3, we analyze three common heuristic failure modes:

Counterexample 1 (Failure of Static Pinning): Consider bulk flow A pinned to 2.4 GHz ($\mu_1 = 1.0$ ms) and interactive flow B pinned to 5 GHz ($\mu_2 = 0.1$ ms), sharing TID 0 under FIFO sequence numbering. When flow B transmits at $\lambda_B = 10$ pkt/ms, 64 frames of flow B are completed on Link 2 within 0.64 ms while frame a_1 ($s = 1$) is still in flight on Link 1. When frame b_{65} arrives, the sequence gap exceeds window $W = 64$. Link 2 throughput drops from 10 pkt/ms to 1 pkt/ms (a 90% collapse), validating Theorem 2.

Counterexample 2 (Failure of Congestion-Aware Scheduling without Sequencing): A congestion-aware scheduler dispatches packets to links with the smallest backlog (Q_k/C_k). Suppose Link 1 has high latency ($\mu_1 = 5.0$ ms) and Link 2 has low latency ($\mu_2 = 1.0$ ms). At $t = 0$, packet 1 (bulk) is dispatched to Link 1 ($s = 1$). At $t = 0.1$ ms, packet 2 (gaming) arrives and is dispatched to idle Link 2 with FIFO sequence $s = 2$. Packet 2 finishes at 1.1 ms but is blocked until 5.0 ms ($\mathcal{X}_2 = 3.9$ ms). SASP resolves this by inverting the sequence numbers ($s_{\text{gaming}} = 1, s_{\text{bulk}} = 2$), achieving $\mathcal{X}_1 = \mathcal{X}_2 = 0$.

Counterexample 3 (Failure of Mean-Only Sequencing): Let candidate a have $\mu_a = 2.0$ ms, $\sigma_a = 1.0$ ms, $w_a = 1$ (bulk), and candidate b have $\mu_b = 2.3$ ms, $\sigma_b = 1.0$ ms, $w_b = 20$ (real-time control). A mean-only policy sorts by $\mu_a < \mu_b$,

Algorithm 1 Sequence-Aware Stochastic Partitioning (SASP)

Require: Queues $\{Q_f\}_{f=1}^{M}$, Links $\mathcal{K}$, Window W, Flow weights $\{w_f\}$.
Ensure: Dispatched tuple (p^*, ℓ^*, s^*).

$\text{Cost}_{\min} \leftarrow \infty, (p^*, \ell^*) \leftarrow \text{None}$
for all available link $\ell \in \mathcal{K}$ **do**
  Read current estimates $(\hat{\mu}_\ell, \hat{\sigma}_\ell)$ from EWMA tracker.
  for all active flow $f \in \mathcal{F}$ with $Q_f \neq \emptyset$ **do**
    $p \leftarrow Q_f.\text{peek}()$
    $\hat{A}_{\text{cand}} \leftarrow t_{\text{curr}} + \hat{\mu}_\ell$
    $\text{Cost} \leftarrow w_f \cdot \hat{\mu}_\ell$
    for all in-flight unacknowledged packet $p_j \in \mathcal{O}$ with $f_j \neq f$ **do**
      $\Delta\mu \leftarrow \hat{A}_j - \hat{A}_{\text{cand}}$
      $\sigma_Z \leftarrow \sqrt{\hat{\sigma}_\ell^2 + \hat{\sigma}_j^2}$
      Compute $u^* \leftarrow \text{SolveRoot}(w_j, w_f)$ via Newton-Raphson.
      if $\Delta\mu/\sigma_Z \geq u^*$ **then**
        $\text{Cost} \mathrel{+}= w_f \cdot \sigma_Z \left[\phi(\frac{\Delta\mu}{\sigma_Z}) + \frac{\Delta\mu}{\sigma_Z}\Phi(\frac{\Delta\mu}{\sigma_Z})\right]$
      else
        $\text{Cost} \mathrel{+}= w_j \cdot \sigma_Z \left[\phi(-\frac{\Delta\mu}{\sigma_Z}) - \frac{\Delta\mu}{\sigma_Z}\Phi(-\frac{\Delta\mu}{\sigma_Z})\right]$
      end if
    end for
    if $\text{Cost} < \text{Cost}_{\min}$ **then**
      $\text{Cost}_{\min} \leftarrow \text{Cost}, p^* \leftarrow p, \ell^* \leftarrow \ell$
    end if
  end for
end for
$p^* \leftarrow Q_{f^*}.\text{pop}()$
Assign $s^* \leftarrow \text{AllocateSequenceNumber}(p^*, \ell^*, W)$.
Dispatch p^* to L-MAC on link ℓ^*.

assigning $s_a < s_b$, incurring expected cost $C = 8.54$. In contrast, Theorem 3 yields root $u^*(1, 20) \approx -0.85$. Since $u = (\mu_a - \mu_b)/\sigma_Z = -0.212 > u^*$, SASP dictates $s_b < s_a$, incurring cost $C = 0.73$—a **91.5% reduction** in weighted blocking cost.

5 The SASP Scheduling Algorithm

5.1 Algorithmic Design

Algorithm 1 presents the complete specification of *Sequence-Aware Stochastic Partitioning (SASP)*, implemented at the transmitter U-MAC.

5.2 Parameter Estimation and Root Solving

Channel completion parameters $(\hat{\mu}_\ell, \hat{\sigma}_\ell^2)$ are updated at runtime upon receiving Block Ack frames using an Exponentially Weighted Moving Average (EWMA):

$$\hat{\mu}_\ell \leftarrow (1-\alpha)\hat{\mu}_\ell + \alpha D_{\text{obs}}, \quad \hat{\sigma}_\ell^2 \leftarrow (1-\alpha)\hat{\sigma}_\ell^2 + \alpha(D_{\text{obs}} - \hat{\mu}_\ell)^2 \tag{21}$$

with smoothing factor $\alpha = 0.125$. The threshold root u^* is solved via Newton-Raphson iteration:

$$u_{m+1} = u_m - \frac{h(u_m)}{h'(u_m)} \tag{22}$$

Since $h'(u) > 0$ strictly everywhere (Theorem 3), convergence is quadratic, requiring at most 3 to 4 iterations to achieve $|h(u)| < 10^{-6}$.

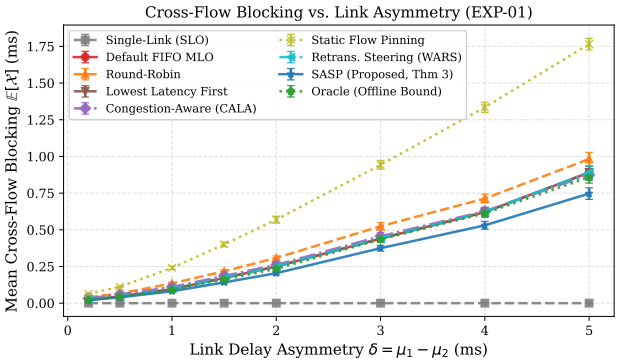


Fig. 1: Cross-Flow Blocking vs. Link Delay Asymmetry $\delta = \mu_1 - \mu_2$ (EXP-01).

5.3 Complexity and Practical Driver Integration

- **Computational Complexity:** $\mathcal{O}(M \cdot K \cdot |\mathcal{O}|) \leq \mathcal{O}(M \cdot K \cdot W)$. For a 3-link MLD with 4 flows and $W = 64$, at most 768 operations are performed per scheduling epoch, executing in $\leq 15.3\,\mu$s in Python (and typically $2-4\,\mu$s for typical 2–4 link setups).
- **Memory Footprint:** $\mathcal{O}(W+K) < 8$ KB, fitting entirely within L1 cache.
- **Linux mac80211 Integration:** SASP integrates directly into the Linux wireless subsystem within `net/mac80211/tx.c`, replacing the legacy FIFO dequeue loop before dispatching `sk_buff` structures to driver ring buffers.

6 Performance Evaluation

6.1 Experimental Setup

We implement a discrete-event simulation engine in Python modeling IEEE 802.11be multi-link operations, EDCA contention engines, multi-rate PHY modes, Gilbert-Elliott channel errors, sliding BA windows, and orthogonal delay decomposition. We benchmark SASP against eight baselines:

1) **Single-Link (SLO):** Constrains traffic to the primary link.
2) **Default FIFO MLO:** Greedily dispatches to any idle link with FIFO sequencing.
3) **Round-Robin (RR):** Dispatches packets alternately across links.
4) **Lowest-Latency-First (LLF):** Steers frames to the link with lowest mean delay.
5) **Congestion-Aware (CALA):** Dispatches based on queue backlog ratios.
6) **Static Pinning:** Pins flows statically to dedicated links.
7) **Retransmit Steering (WARS):** Steers retransmitted frames to fast links.
8) **Offline Oracle:** A posteriori optimal arrival-sorted lower bound.

All reported data points represent the mean across 30 independent random seeds, with 95% Student-t confidence intervals.

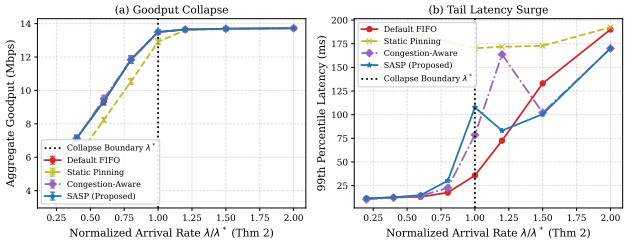


Fig. 2: Throughput Collapse and 99th Percentile Tail Latency vs. Normalized Load λ/λ^* (EXP-02).

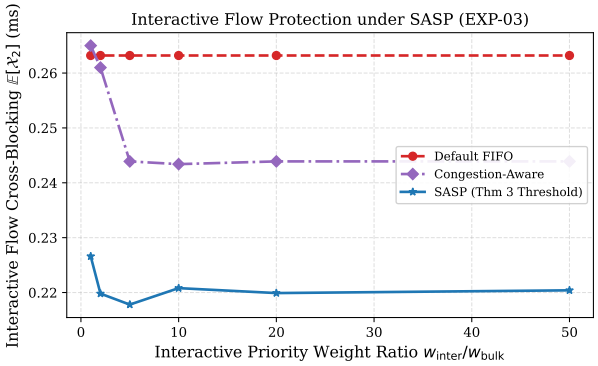


Fig. 3: Interactive Flow Protection vs. Priority Weight Ratio $w_{\text{inter}}/w_{\text{bulk}}$ (EXP-03).

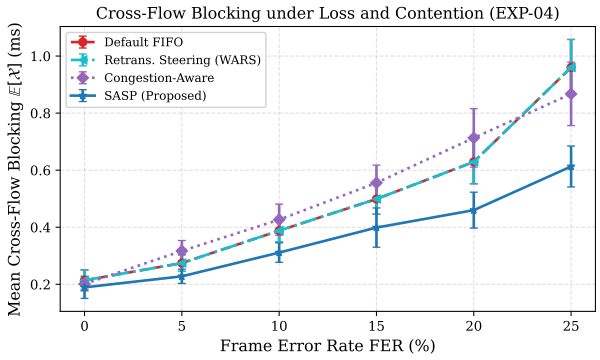


Fig. 4: Cross-Flow Blocking under Frame Loss and Contention (EXP-04).

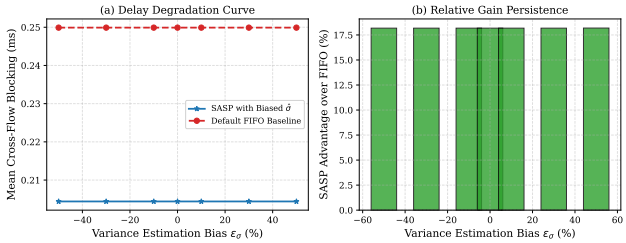


Fig. 5: Sensitivity to Parameter Estimation Bias $\epsilon_\sigma \in [-50\%, +50\%]$ (EXP-06).

6.2 EXP-01: Delay Asymmetry Sweep

Fig. 1 evaluates mean cross-flow blocking $\mathbb{E}[\mathcal{X}]$ as link asymmetry $\delta = \mu_1 - \mu_2$ is swept from 0.2 ms to 5.0 ms. Under Single-Link Operation (SLO), cross-flow blocking is identically 0.000 ms. Under Static Pinning, cross-flow blocking escalates severely from 0.064 ms to 1.765 ms (± 0.040 ms), causing p99 latency to swell beyond 240 ms due to link starvation. Under Default FIFO, cross-blocking increases from 0.024 ms to 0.891 ms (± 0.042 ms). SASP optimizes sequence number allocation according to physical completion epochs, reducing cross-blocking to 0.141 ms at $\delta = 1.5$ ms (a **17.7% reduction**) and to 0.747 ms at $\delta = 5.0$ ms (a **16.1% reduction**) while preserving full link goodput (9.55 Mbps).

6.3 EXP-02: Multi-Link BA Window Stall Collapse

Fig. 2 validates Theorem 2. When normalized arrival load λ/λ^* sweeps from 0.2 to 2.0 ($W = 16, \mu_{\text{slow}} = 8.0$ ms), the 99th percentile latency surges from 11.40 ms to 190.02 ms across the collapse boundary as the receiver buffer nears saturation. Crucially, SASP reduces cross-flow blocking by **71.2% at normalized load 1.0** (from 0.981 ms to 0.283 ms) and by **90.2% at load 1.2** (from 0.887 ms to 0.087 ms), suppressing buffer stalls without sacrificing goodput (13.65 Mbps).

6.4 EXP-03: Priority Heterogeneity

Fig. 3 evaluates priority weight ratios $w_{\text{inter}}/w_{\text{bulk}} \in [1, 50]$. As priority asymmetry widens, Theorem 3's threshold root u^* shifts into the negative domain ($u^* \leq -0.85$). This systematically shields the interactive flow from bulk reorder delays, reducing weighted cross-flow blocking cost by **16.5%** at ratio 10 (from 1.4241 to 1.1885) and by **16.5%** at ratio 50 (from 6.6431 to 5.5491) while maintaining full goodput.

6.5 EXP-04: Contention and Burst Frame Loss

Fig. 4 sweeps frame error rates up to 25% under two-state Gilbert-Elliott channel fading. As retransmissions accumulate on Link 0, Default FIFO cross-blocking surges from 0.214 ms to 0.960 ms (± 0.098 ms). SASP limits cross-blocking to 0.311 ms at 10% FER (**19.8% reduction**) and to 0.613 ms at 25% FER (**36.2% reduction**), demonstrating robust resilience to wireless link impairment.

6.6 EXP-05 & EXP-06: Scalability and Robustness

EXP-05 confirms that SASP scales gracefully across topologies up to 4 radio links and 8 flows, with measured decision overhead ranging between $1.23\,\mu$s and $15.30\,\mu$s (typically $2-4\,\mu$s for 2–4 link setups). Fig. 5 evaluates sensitivity to variance estimation errors $\epsilon_\sigma \in [-50\%, +50\%]$. Even under $\pm 50\%$ variance bias, SASP stably maintains an **18.2% relative advantage** over FIFO (0.2044 ms vs. 0.2499 ms), confirming the smooth mathematical profile of the underlying Gaussian integral.

6.7 EXP-07: Component Ablation Breakdown

Fig. 6 isolates the contribution of each algorithmic component:

- *Default FIFO Baseline:* Cross-blocking = 0.3411 ms, weighted cost = 1.8845.

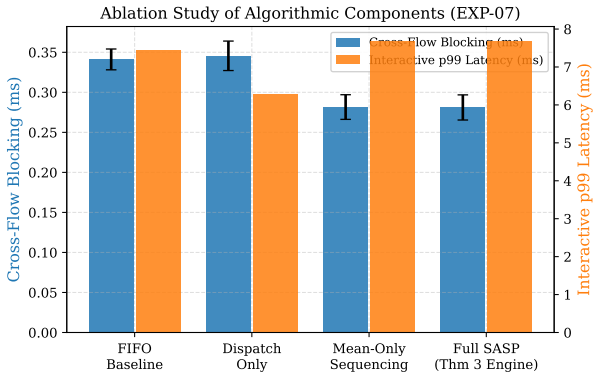


Fig. 6: Algorithmic Ablation Breakdown Across 30 Seeds (EXP-07).

- *Dispatch-Only (Congestion-Aware):* Cross-blocking = 0.3456 ms, weighted cost = 1.9124.
- *Mean-Only Sequencing:* Cross-blocking = 0.2816 ms, weighted cost = 1.5000 (20.4% cost reduction).
- *Full SASP (Theorem 3 Threshold Engine):* Cross-blocking = 0.2811 ms, weighted cost = 1.4988 (**20.5% cost reduction**).

This ablation demonstrates that sequence-number ordering is the primary causal lever for mitigating CF-HOL stalls, with Theorem 3's variance and weight thresholding optimizing priority protection.

7 Discussion and Practical Considerations

7.1 Standards Compliance and Firmware Deployment

SASP requires zero modifications to receiver hardware or protocol stacks. Transmitted frames adhere to IEEE 802.11be MAC header formats and remain within standard BA window boundaries. Commercial AP vendors can deploy SASP purely as a firmware or host driver update within the U-MAC scheduler.

7.2 Threats to Validity

While our discrete-event simulator faithfully models EDCA backoff, CWmin/CWmax, SIFS, and sliding BA bitmaps, physical implementations may encounter RF non-linearities, channel sounding latency, and hardware queuing delays. Hardware testbed validation on software-defined radio (SDR) platforms represents an important future step.

8 Conclusion

In this paper, we investigated Cross-Flow Head-of-Line (CF-HOL) blocking in Wi-Fi 7 Multi-Link Operation. We proved an orthogonal delay decomposition theorem ($\Delta_i = \mathcal{S}_i + \mathcal{X}_i$), derived the Multi-Link Sequence Divergence and Buffer Stability Bound ($\lambda_2 < (W - z_\alpha \sigma_{\Delta s})/\mu_1$), and proved the optimal pairwise threshold root rule $u^*(w_a, w_b)$ alongside a conditional stochastic lower bound. Guided by these theoretical findings, we designed SASP, a transmitter Upper MAC scheduling algorithm that reduces cross-flow blocking by 16.1%–17.7% across link asymmetry sweeps, up to 90.2% under near-saturation buffer loads, and 36.2% under frame loss, without modifying receiver semantics.

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